

SOGO-HAYA, Russian Federation

WMO: 245570

Lat: 64.283N

Lon: 126.450E

Elev: 279

StdP: 14.55

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-57.3	-54.3	-60.7	0.1	-56.9	-57.7	0.2	-54.6	17.9	-24.5	16.0	-28.4	2.2	110	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	83.7	66.6	79.8	64.7	76.0	62.6	69.0	79.8	66.5	76.0	64.2	73.4	4.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.7	73.4	62.8	86.2	72.0	60.4	79.2	69.2	33.4	79.4	31.4	75.8	29.6	73.4	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.1	13.8	-59.6	89.5	4.1	3.3	-62.5	91.8	-64.9	93.7	-67.2	95.6	-70.1	98.0		
(5)			-59.6	72.7	4.1	3.6	-62.5	75.3	-64.9	77.4	-67.2	79.4	-70.1	82.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	-37.0	-24.9	-0.1	21.9	41.6	59.0	64.2	58.2	42.3	17.8	-18.4	-33.0		
	DBStd	37.62	11.47	10.84	14.04	11.06	9.82	7.68	7.29	6.70	7.98	10.78	12.46	12.54		
	HDD50	13392	2697	2097	1552	842	299	14	2	11	254	999	2052	2573		
	HDD65	17940	3162	2517	2017	1292	727	209	105	227	680	1464	2502	3038		
	CDD50	1050	0	0	0	0	38	283	440	265	24	0	0	0		
	CDD65	123	0	0	0	0	0	28	79	16	0	0	0	0		
	CDH74	1247	0	0	0	0	6	345	754	141	1	0	0	0		
	CDH80	313	0	0	0	0	0	72	226	15	0	0	0	0		
	WSAvg	5.7	4.9	5.9	5.9	6.4	5.9	5.7	5.9	5.5	5.3	6.0	5.4	5.0		
	PrecAvg	14.3	0.5	0.4	0.5	0.7	1.4	1.7	2.1	2.1	1.8	1.7	0.9	0.6		
(15) Precipitation	PrecMax	20.6	1.1	1.2	1.4	1.5	2.8	3.6	4.3	5.1	3.8	3.1	2.2	1.2		
	PrecMin	10.1	0.2	0.0	0.1	0.2	0.2	0.5	0.7	0.5	0.2	0.2	0.4	0.2		
	PrecStd	2.1	0.2	0.3	0.3	0.3	0.6	0.7	0.9	0.9	0.9	0.6	0.4	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.1	5.4	34.2	49.0	73.5	85.6	90.4	81.9	69.8	40.7	17.9	-1.9		
		MCWB	-3.7	4.3	29.7	39.3	56.2	64.7	71.3	65.2	57.6	38.3	15.4	-3.1		
	2%	DB	-10.3	1.9	29.4	44.9	67.8	82.1	85.7	77.8	63.9	37.2	11.6	-6.9		
		MCWB	-10.9	1.1	25.9	35.8	51.2	64.4	68.6	63.7	53.7	34.9	10.5	-7.6		
	5%	DB	-16.3	-4.8	25.5	41.7	62.7	78.0	81.9	73.9	58.7	34.1	6.3	-10.0		
		MCWB	-16.7	-5.6	22.2	33.8	49.5	62.2	66.5	62.4	50.3	32.0	5.6	-10.5		
	10%	DB	-21.2	-9.4	21.9	38.7	57.8	73.9	77.9	70.2	54.5	31.3	-0.9	-14.6		
		MCWB	-21.5	-9.9	19.3	32.0	46.7	60.6	64.6	60.9	47.7	29.7	-1.5	-15.0		
	0.4%	WB	-3.7	4.4	30.7	40.5	56.4	69.6	73.0	68.9	60.0	39.1	15.7	-2.7		
		MCDB	-3.0	5.1	34.0	47.8	70.4	81.4	85.7	77.8	68.7	39.7	17.5	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-10.7	0.9	26.1	36.8	52.8	65.7	70.5	65.7	54.5	35.7	10.6	-7.4		
		MCDB	-10.0	1.8	29.0	42.9	65.1	77.5	82.4	73.0	61.0	37.0	11.5	-6.9		
	5%	WB	-16.6	-5.6	22.5	34.8	50.0	63.4	68.4	63.7	51.9	32.3	5.5	-10.3		
		MCDB	-16.3	-4.9	25.1	40.3	61.9	75.4	79.1	71.3	57.7	33.8	6.2	-9.8		
	10%	WB	-21.4	-10.1	19.2	32.6	47.0	61.5	66.2	61.7	48.7	29.8	-1.4	-15.0		
		MCDB	-21.2	-9.6	22.0	38.4	56.9	71.9	75.3	69.0	53.4	31.3	-0.8	-14.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	12.9	17.9	19.1	16.5	18.9	17.7	17.1	14.7	9.1	10.0	9.0		
		MCDBR	11.7	13.5	19.0	19.2	22.5	26.2	23.2	23.3	21.3	9.7	10.5	12.6		
		MCWBR	11.6	12.9	15.9	12.7	12.4	12.3	9.9	11.1	11.8	7.2	10.3	12.4		
		MCDBR	11.8	13.9	18.4	17.1	21.6	24.4	20.4	19.6	20.1	8.4	10.3	12.3		
	5% WB	MCWBR	11.6	13.4	15.7	11.6	12.2	12.6	10.3	10.3	12.1	6.8	10.2	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.189	0.245	0.290	0.338	0.387	0.407	0.485	0.469	0.337	0.266	0.187	0.179		
	taud		1.902	2.082	2.091	2.090	2.221	2.292	2.044	2.035	2.406	2.258	1.910	1.886		
	Ebn at Noon		117	211	249	261	259	256	227	216	236	203	112	44		
	Edh at Noon		9	20	31	39	38	36	46	42	23	16	8	3		
(44) All-Sky Solar Radiation	RadAvg		61	336	918	1581	1708	1866	1699	1257	749	297	104	22		
	RadStd		3	16	43	68	157	130	85	110	103	39	8	1		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	-292	N/A	N/A	

Nomenclature: See separate page